

Country Availability Check

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the application's homepage or the country selection/registration page.
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+ Acceptance Criteria:

- As a client user I will be able to access the Practice Software Testing website, then verify country availability for registration:
 - As a client user it is mandatory to select 'Country' from the deployed list of countries configured in 'Managed Countries' configuration in Admin Portal.
 - As a web application the 'Country' drop down list is provided for countries at which this website can be utilized.
- As a web application, when the user accesses the site, it will validate the availability of the application in the user's current country via IP auto-detection.
- The website will grant full access if the country is supported, and automatically select it in the 'Country' dropdown.

- As a web application, if the detected country is unsupported, the user cannot proceed in the application. An info message will be displayed to inform the client user that the service is unavailable in their region.
 - As a web application, during registration the selected country "Terms & Conditions" will be displayed for review by client users to accept or not.
 - The client user should click Agree so that the system can proceed with the registration. In case the client user pressed Disagree so the user cannot proceed in the application.
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+ Expected Results:

- This client user's location is verified against the supported countries list, and if supported, the user is permitted to access the website and proceed to the registration page.
- In case the client user's country is not supported, a relevant info message will be displayed to inform the user that the service is unavailable, and they cannot proceed.
- The website database will record the selected country for this client user to be used during the registration process and to display the correct products and pricing.

- The user is redirected to Accept the 'Terms & Conditions' set for this website at the selected country.
 - In case the client user did not accept terms & conditions, a relevant info message will be displayed to inform the user that s/he cannot proceed with registration.
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+ Dependencies:

- Registration Data: According to selected 'Country', the website will select the relevant 'Country' at the registration data.

User Registration

+ Prerequisites:

- The user is connected to the internet.
 - The user is on the Registration page.
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+ Acceptance Criteria:

- As a client user, I will be able to access the registration page.
- As a client user, I will be able to enter required data such as (Username, Email, Password, Confirm Password, Phone, etc.).
- As a web application, all required fields must be filled, otherwise validation messages will be displayed.
- As a web application, the system should validate the email format.
- As a web application, the system should validate that the password meets the required criteria (e.g., minimum length, characters).
- As a web application, the system should ensure that Password and Confirm Password match.
- As a web application, the system should check that the email is not already registered.
- As a web application, if the email already exists, an error message will be displayed: "Email already exists".

- As a client user, I must accept the Terms & Conditions before completing registration.
 - As a web application, if the user does not accept Terms & Conditions, the system will not allow registration.
 - As a web application, upon successful registration, the user account will be created and stored in the database.
 - As a web application, the user will be redirected to the login page or homepage after successful registration.
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+ Expected Results:

- The user successfully registers when valid data is entered.
 - The system prevents registration when invalid or missing data is provided.
 - The system ensures that duplicate accounts are not created.
 - The user data is stored securely in the database.
 - The user can log in using the registered credentials.
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+ Dependencies:

- **Database Availability:** According to the system status, the database must be available to store user data.

- **Validation Services:** According to the system configuration, email/validation services must be working properly.
- **Terms Configuration:** According to the legal requirements, Terms & Conditions must be configured in the system.

User Login

+ Prerequisites:

- The user is connected to the internet.
 - The user has already registered an account.
 - The user is on the Login page of the website.
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+ Acceptance Criteria:

- As a client user, I will be able to access the login page and enter my credentials (Email & Password).
- As a client user, I must enter a valid registered email and correct password to log in successfully.
- As a web application, the system will validate the entered credentials against the database.
- As a web application, if the credentials are correct, the user will be redirected to the homepage/dashboard.
- As a web application, if the email or password is incorrect, an error message will be displayed: "Invalid email or password".
- As a web application, the password field should be masked for security purposes.
- As a web application, the system should not allow login if any required field is empty, and a validation message should be displayed.
- As a client user, I can click on "Forgot Password" to reset my password.

+ Expected Results:

- The user successfully logs in when valid credentials are provided.
 - The system prevents login when invalid credentials are entered and shows an appropriate error message.
 - The system ensures that user data is securely handled during login.
 - The user is redirected to the correct page after successful login.
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+ Dependencies:

- User account must exist in the database.
- Database connection must be active.
- Authentication service must be working properly.

User Forgot Password

+ Prerequisites:

- The user is connected to the internet.
 - The user has a registered account.
 - The user is on the Login page.
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+ Acceptance Criteria:

- As a client user, I will be able to click on "Forgot Password" from the login page.
- As a client user, I will be asked to enter my registered email address.
- As a web application, the system will validate if the entered email exists in the database.
- As a web application, if the email is valid, a password reset link will be sent to the user's email.
- As a web application, if the email does not exist, an error message will be displayed: "Email not found".
- As a client user, I can click on the reset link received in my email.
- As a web application, the reset link should be time-limited for security purposes.
- As a client user, I will be able to enter a new password and confirm it.
- As a web application, the system should validate that the new password meets the required criteria.

- As a web application, the system should ensure that the password and confirm password fields match.
 - As a web application, after successful reset, the user will be redirected to the login page.
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+ Expected Results:

- The user receives a password reset email when a valid email is entered.
 - The user is able to reset the password successfully using the reset link.
 - The system prevents reset when invalid or unregistered email is entered.
 - The system ensures password security and validation during reset process.
 - The user can log in using the new password after reset.
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+ Dependencies:

- User email must exist in the database.
- Email service must be working properly.
- Reset password functionality must be active.

Customer Profile Management

+ Prerequisites:

- The user is logged into the website.
 - The user has accessed the "Profile" or "Account Settings" page.
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+ Acceptance Criteria:

- As a logged-in user, I will be able to update my profile information, including First name, last name, phone, and address fields.
- As a web application, the system shall allow the user to reset their password.
- As a web application, the password reset process must ensure that the user cannot reuse their current password.
- Upon a successful password reset, the system will automatically log the user out from the current session.

+ Expected Results:

- The user's updated profile details (First name, last name, phone, address) are saved in the database and remain correct after a page reload.
 - Any attempt to reset the password using the current password will be rejected by the system.
 - After a successful password reset, the user session state changes to "logged out," and the user must log in again using the new password.
 - The system validates input fields based on Equivalence Partitioning (EP) for phone numbers and postal codes, and Boundary Value Analysis (BVA) for character limits (e.g., max 40 characters).
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+ Dependencies:

- **Authentication Module:** The user must maintain an active session to access and modify profile data.
- **Session Management:** The system must be able to trigger a global logout across the application after a password change.

Customer Favorites

+ Prerequisites:

- The user is connected to the internet.
 - The user is logged into the website as a customer.
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+ Acceptance Criteria:

- As a logged-in customer, I will be able to add a product to favorites from the product listing page or the detail page.
 - As a web application, the favorited products must appear accurately in the "Favorites list".
 - As a web application, the user can remove a favorite from the list, and it should no longer appear.
 - As a web application, if a user attempts to favorite a product while logged out, a proper error message should be displayed.
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+ Expected Results:

- When adding multiple products, all selected items are displayed in the favorites list and persist even after logging out and back in.

- When a product is removed, the system confirms only that specific product is deleted, and the state changes from 'favorite' to 'not favorite'.
 - Any attempt to add the same product twice should be handled by the system to prevent duplicates or show a relevant error.
 - The website database will record the favorited products for this client user to ensure they are available across different sessions.
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+ Dependencies:

- **Authentication Module:** The ability to save favorites is dependent on the user's active session and account ID.
- **Product Catalog:** Favorites data is linked to the existence of valid products in the database.

Customer Invoices

+ Prerequisites:

- The user is connected to the internet.
 - The user is successfully logged into the application.
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+ Acceptance Criteria:

- As a logged-in user, I will be able to access the "Invoices" section within my account settings.
- As a web application, the system shall display a list of all completed orders with their corresponding invoice details.
- As a web application, the system shall provide a download button/link for each invoice in a standard format (e.g., PDF).
- As a web application, the invoice must contain accurate details including Order ID, Date, Product List, Tax, and Total Amount.

+ Expected Results:

- The system correctly lists all invoices associated with the user's account ID.
 - When the user clicks "Download", the invoice file is generated accurately and saved to the user's device.
 - All financial calculations (Subtotal, Tax, Discounts, Total) within the invoice match the actual order data in the database.
 - The system prevents users from accessing or downloading invoices belonging to other accounts.
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+ Dependencies:

- **Order History Module:** The invoices are generated based on the existing order records in the database.
- **Billing Service:** The system must have an active service to generate and format the invoice documents (e.g., PDF generator).

Customer Messages

+ Prerequisites:

- The user is connected to the internet.
 - The user is logged into the application.
 - A previous contact message or conversation thread exists between the customer and support.
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+ Acceptance Criteria:

- As a customer, I will be able to view all past contact messages, which must appear in chronological order.
- As a web application, a reply field shall be available for the user to type and send messages.
- The system must ensure that messages are sent successfully and are linked to the correct conversation thread.

+ Expected Results:

- Upon opening a message thread, the user can verify that the conversation history is ordered correctly by time/date.
 - When a reply is sent, it appears instantly in the chat window or immediately after a page refresh.
 - Any attempt to send an empty reply will be rejected, and the system will display a validation error.
 - If the user attempts to send a reply while offline, the application should handle the error gracefully and notify the user.
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+ Dependencies:

- **Customer Support Portal:** The system must be integrated with the support backend to retrieve and send messages.
- **Notification System:** (Optional) The system may depend on a notification module to alert the user of new replies from support.

Account Lockout Policy

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the application's login page.
 - The user has a pre-existing registered account in the system.
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+ Acceptance Criteria:

- As a web application, the system will monitor the number of failed logins attempts for each unique email address.
- It is mandatory for the system to trigger the "Account Lockout" status after 5 consecutive failed attempts with an incorrect password.
- As a web application, when the lockout is triggered, the system will automatically display an info message to the user stating that the account is temporarily disabled for security reasons.
- The system will grant access again only after a 30-minute cooling period has elapsed from the time of the last failed attempt.
- As a security measure, during the lockout period, the system must prevent any login attempts even if the correct password is provided.

- The website will record the logout event in the database, including the timestamp and user IP, to be used for security auditing.
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+ **Expected Results:**

- The account status will be updated to "Locked" in the database, and the user will be prohibited from proceeding in the application.
 - A relevant info message will be displayed to inform the user that they cannot proceed due to multiple failed attempts.
 - The user will be redirected to a security recovery page (or Password Reset) to re-verify his/her identity and regain access.
 - Full access will only be granted once the account is unlocked or the user successfully completes the authentication/reset process.
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+ **Dependencies:**

- **Account Status:** According to the login attempts, the website will update the relevant 'Status' in the user's registration data.
- **Email Notification Service:** The system must be integrated with an SMTP server to notify the user of the logout.

Multi-Factor Authentication (MFA)

+ Prerequisites:

- The user is connected to the internet.
 - The user is logged into their account and has accessed the security or profile settings.
 - The user has a TOTP-compatible authenticator app installed on their mobile device.
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+ Acceptance Criteria:

- As a web application, the system will provide a unique QR code for the client user to scan using an authenticator app to enable MFA.
- It is mandatory for the system to verify the Time-based One-Time Password (TOTP) entered by the user before completing the activation process.
- As a web application, once MFA is enabled, the system will prompt the user for a valid TOTP during every subsequent login attempt.
- The system will grant full access to the application only after the user provides the correct 6-digit TOTP.
- In case the user enters an invalid TOTP, the system will display a **relevant info message** and they cannot proceed to the dashboard.

+ Expected Results:

- The website database will record the MFA status as "Enabled" for this client user to be used during the authentication process.
 - Upon a successful login attempt with correct credentials, the user is redirected to the MFA verification page to enter the received TOTP.
 - If a valid TOTP is provided, the user is permitted to access the website and proceed to the application.
 - In case the client user provides an incorrect TOTP, a relevant info message will be displayed to inform the user that s/he cannot proceed.
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+ Dependencies:

- **TOTP Verification:** The system must synchronize the server-side time with the TOTP generation logic to ensure accurate code validation.
- **Authentication Data:** According to the MFA status, the website will select the relevant authentication flow during the login process.

Contact Support Form

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the application's contact us page or support portal.
 - The user is logged into their account to infer profile data.
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+ Acceptance Criteria:

- As a web application, it is mandatory to fill the 'Name' (max 120 chars), 'Subject' (max 120 chars), and 'Message' (max 250 chars) fields before submission.
- The system will automatically infer the 'Email' from the user profile; if not available, the 'Email' field becomes mandatory to proceed.
- As a web application, the file upload functionality must restrict attachments to txt, pdf, and jpg formats only.
- It is mandatory for the system to validate file size, rejecting any file that is 0KB or exceeds 500KB.
- In case of invalid file types or sizes, a relevant info message will be displayed to inform the client user of the specific error.

+ Expected Results:

- This client user's input is validated against the character limits and file requirements, and if valid, the user is permitted to submit the support request.
 - The website database will record the submitted message, attachment, and user details to be used for support tracking and resolution.
 - In case the user uploads an unsupported file type, a relevant info message will be displayed to inform the user that s/he cannot proceed.
 - The user is redirected to a 'Success' page or shown a confirmation message upon successful submission of the contact form.
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+ Dependencies:

- **User Profile Data:** According to the login status, the website will select the relevant 'Email' at the contact form data.
- **File Configuration:** According to the system's security policy, the website will update the relevant 'Upload Restrictions' to prevent unauthorized file execution.

Product Filtering and Sorting

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the application's homepage or the product listing page.
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+ Acceptance Criteria:

- As a web application, the system will provide a 'Filter' section where selecting a specific category or brand shows matching products only.
- It is mandatory for the system to support sorting functionality by 'Name' and 'Price' in both ascending and descending directions.
- As a web application, the system must validate the 'Price Range' input and display products that fall within the specified range correctly.
- The system will provide a 'Reset' functionality to clear all active filters and search queries to display the full product list again.
- In case no products match the selected criteria, a relevant info message will be displayed to inform the user that no results were found.

+ Expected Results:

- The client user's selection is verified against the product database, and if matching products exist, the user is permitted to view the filtered list.
 - The website database will record the active search/filter parameters for the current session to maintain the correct product display.
 - In case the 'Reset' button is clicked, all filters are cleared automatically, and the user is redirected to the full product catalog.
 - Combined filters (Category + Sort + Price) will be processed simultaneously, and the website will select the relevant items to display.
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+ Dependencies:

- **Product Data:** According to the selected 'Filters', the website will select the relevant products at the inventory database.
- **Price Configuration:** According to the user's currency and region, the website will display the correct products and pricing during the filtering process.

Category Page

+ Prerequisites:

- The user is connected to the internet.
 - The user is on the Home page of the website.
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+ Acceptance Criteria:

- As a client user, I will be able to see all available categories (Hand Tools, Power Tools, etc.) on the homepage or side menu.
- As a client user, I can click on any category to view its specific products.
- As a web application, the system should display the selected category name as a title on the page.
- As a web application, the system should allow filtering products by price and brand.
- As a web application, the system should provide sorting options (e.g., price low to high, name A-Z).
- As a web application, if no products are available in a category, a message should be displayed:” No products found”.
- As a client user, I can see the product image, name, and price for each item in the list.

+ Expected Results:

- The system displays the correct products for each selected category.
 - Filters and sorting work properly and update the product list.
 - The user is able to navigate between different categories easily.
 - All product information (Price, Image, Title) is visible and correct.
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+ Dependencies:

- Categories and products must be set up in the database.
- Database connection must be active.
- Filtering and sorting logic must be implemented.

Product Detail Page

+ Prerequisites:

- The user is connected to the internet.
 - The user has navigated to a specific product page.
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+ Acceptance Criteria:

- As a client user, I will be able to see the full product details including (Product name, Price, Description, and Image).
- As a client user, I will be able to see if the product is "In Stock" or "Out of Stock".
- As a client user, I can choose the quantity of the product I want to buy.
- As a client user, I can click on "Add to cart" button to add the product to my shopping cart.
- As a web application, the system should show a success message when the product is added to the cart.
- As a web application, the system should update the cart icon with the number of items added.
- As a web application, if the product is out of stock, the "Add to cart" button should be disabled.

+ Expected Results:

- The user can see all product information correctly.
 - The product is added to the cart successfully with the correct quantity.
 - The system prevents adding products that are out of stock.
 - The cart counter updates correctly after adding the product.
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+ Dependencies:

- Product data must be available in the database.
- The shopping cart functionality must be active.
- Database connection must be working.

Shopping Cart

+ Prerequisites:

- The user is connected to the internet.
 - The user has added at least one product to the cart.
 - The user is on the Shopping Cart page.
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+ Acceptance Criteria:

- As a client user, I will be able to see all products added to the cart with their name, price, and quantity.
- As a web application, the system should calculate the total price automatically.
- As a client user, I can change the quantity of any product in the cart.
- As a client user, I can delete any product from the cart.
- As a web application, if the cart is empty, a message” Your cart is empty” should be displayed.
- As a client user, I can click on” Proceed to checkout” to complete the purchase.
- As a web application, the system should update the total price immediately when a product is removed or quantity is changed.

+ Expected Results:

- The cart displays the correct products and prices.
 - Total price calculation is accurate.
 - Items are removed or updated successfully.
 - The user can move to the checkout page easily.
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+ Dependencies:

- Products must be added to the cart first.
- Cart functionality and calculations must be working.
- Database connection must be active.

Check Out & Payment

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the website or the shopping cart / checkout page.
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+ Acceptance Criteria:

- As a client user, I will be able to access the Practice Software Testing website, then proceed to the checkout and payment process.
- As a client user, it is mandatory to enter shipping and payment details in order to complete the purchase process.
- As a web application, the "Checkout" page is provided to display order details and complete the payment process.
- As a web application, when the user accesses the checkout page, the system will display the order summary including products and prices.
- As a web application, the user must enter the required shipping details to proceed with the order.
- As a web application, the system will validate the entered data before proceeding.
- As a web application, the user must select an available payment method to complete the process.

- As a web application, upon entering payment details, the system will validate the data.
 - As a web application, when the payment is confirmed, the system will process the payment and create the order.
 - As a web application, if the payment is successful, the user is allowed to complete the order and a confirmation message will be displayed.
 - As a web application, if the payment fails, the user cannot proceed with the order and an informative message will be displayed.
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+ Expected Results:

- The user's data and order are validated, and if valid, the user is allowed to complete the purchase process.
- In case of successful payment, the order is created successfully and a confirmation message is displayed to the user.
- In case of failed payment, an informative message is displayed, and the user cannot proceed.
- The system records the order data in the database for future use.
- The order details are displayed to the user after successful completion.

+ Dependencies:

- **Cart Data:** The selected products in the cart are used during the checkout process.
- **User Data:** Shipping and payment details are used to complete the order.
- **Payment System:** The completion of the process depends on the integrated payment system.

Geolocation Discount

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the Practice Software Testing website.
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+ Acceptance Criteria:

- As a client user, I will be able to access the Practice Software Testing website, then receive location-based discounts if applicable.
- As a client user, it is not mandatory to manually select location, as it can be detected automatically via IP.
- As a web application, the system will detect the user's location using IP auto-detection.
- As a web application, the system will check if any discounts are configured for the detected country or region.
- As a web application, if a discount is available for the user's location, it will be applied automatically to the product prices.
- As a web application, the discounted price will be displayed clearly to the user.
- As a web application, the original price may be displayed along with the discounted price (if applicable).

- As a web application, if no discount is configured for the user's location, the system will display the standard prices.
 - As a web application, the discount logic will be applied consistently across product listing, product details, and checkout pages.
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+ Expected Results:

- The user's location is detected successfully by the system.
 - If a discount is available for the detected location, it is applied automatically.
 - The user sees updated prices reflecting the applied discount.
 - If no discount is available, normal pricing is displayed without errors.
 - The system ensures consistency of pricing across all pages.
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+ Dependencies:

- Geolocation Service: Used to detect user location via IP.
- Discount Configuration: Managed in the system for specific countries or regions.
- Pricing Engine: Responsible for applying and displaying correct prices.

Combined Product Discount

+ Prerequisites:

- The user is connected to the internet.
 - The user has accessed the Practice Software Testing website.
 - The user has added multiple eligible products to the shopping cart.
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+ Acceptance Criteria:

- As a client user, I will be able to access the Practice Software Testing website, then receive a combined product discount when applicable.
- As a client user, it is required to have multiple specific products added to the cart in order to qualify for the combined discount.
- As a web application, the system will identify eligible products that are part of a combined discount rule.
- As a web application, when the required combination of products is present in the cart, the system will automatically apply the combined discount.
- As a web application, the discounted price will be calculated based on the defined discount rules for the combined products.
- As a web application, the system will display the updated total price after applying the combined discount.

- As a web application, the discount should be reflected in the cart summary and checkout page.
 - As a web application, if the required combination of products is not met, no combined discount will be applied.
 - As a web application, the system will continue to display the standard pricing if the discount conditions are not satisfied.
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+ Expected Results:

- The system detects eligible product combinations in the cart.
 - If the required combination is fulfilled, the combined discount is applied successfully.
 - The user sees the updated total price after discount.
 - If the required combination is not fulfilled, no discount is applied and standard pricing is shown.
 - The pricing remains consistent across the cart and checkout pages.
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+ Dependencies:

- Cart Data: Used to identify products added by the user.
- Discount Rules Configuration: Defines eligible product combinations and discount conditions.

- Pricing Engine: Responsible for calculating and applying the combined discount.
- Product Catalog: Used to match products with discount rules.

Admin Login Functionality

+ Prerequisites:

- The user is connected to the internet.
- The user has accessed the login page of the "Practice Software Testing" website (<https://practicesoftwaretesting.com>).
- The user possesses valid Admin credentials (Email and Password).

+ Acceptance Criteria:

- **Accessing Admin Portal:** As an administrator, I must be able to navigate to the login section to access the backend management tools.
- **Mandatory Credentials:** It is mandatory for the user to provide a registered admin email and the corresponding password to gain access.
- **Credential Validation:**
 - The system shall validate the email format (e.g., admin@practicesoftwaretesting.com).
 - The system shall verify if the credentials match the records in the administrative database.

- **Secure Authentication:** The password field must mask the characters (bullets/asterisks) to maintain security during the login process.
- **Session Management:** Upon successful login, the system will grant full access to the Admin Dashboard and maintain the session until the user logs out or the session expires.
- **Error Handling:** If the email or password provided is incorrect, the system must prevent access and display a relevant error message (e.g., "Invalid credentials").

+ **Expected Results:**

- **Successful Login:** Upon entering the correct email (**admin@practicesoftwaretesting.com**) and password (**welcome01**), the user is redirected to the Admin Dashboard.
- **Dashboard Authority:** The administrator will see specific management features such as "Products," "Orders," "Invoices," and "User Management" that are not available to regular clients.
- **Failed Authentication:** In case of incorrect credentials, the user remains on the login page, and an info message is displayed stating that the login attempt failed.

- **Database Verification:** The website database will verify the 'Role' associated with the email to ensure the user has 'Admin' privileges before granting access to protected routes.

+ **Dependencies:**

- **Account Role:** The login functionality depends on the user being assigned the "Admin" role in the system configuration.
- **System Configuration:** The Admin Portal must be correctly configured to recognize the master admin account provided for management tasks.